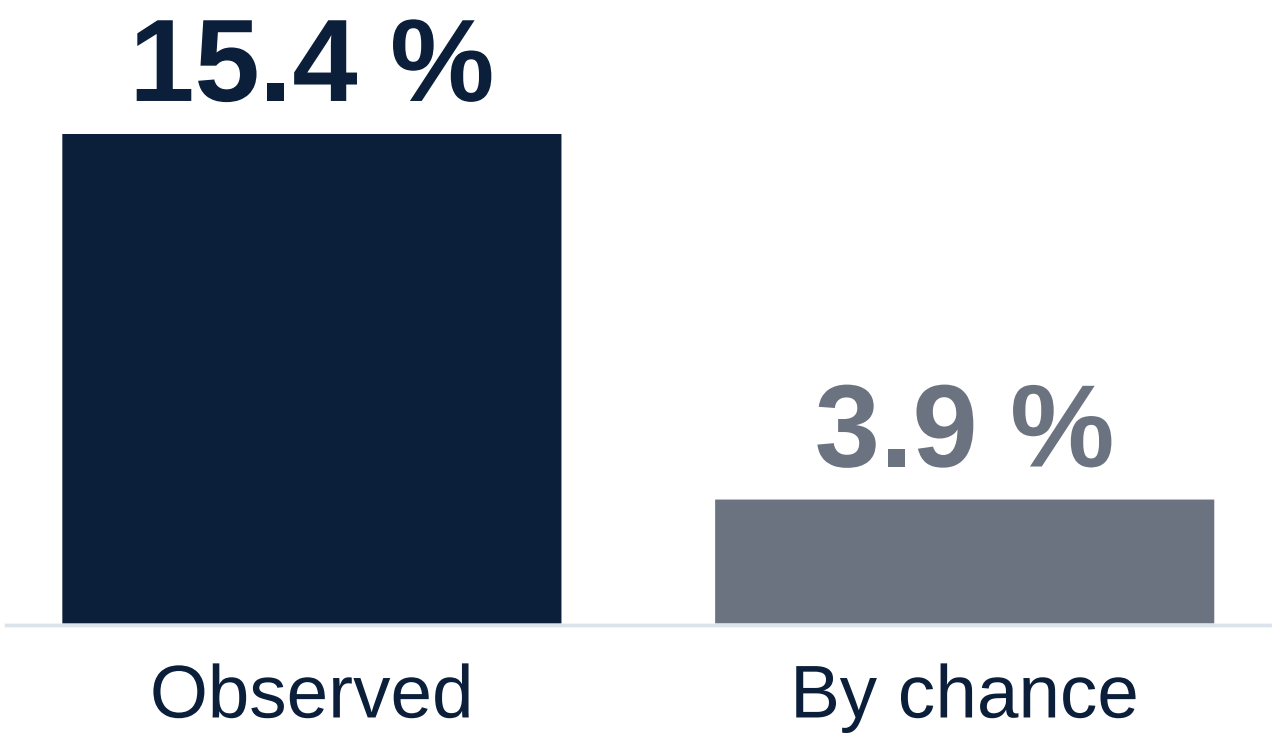
May 2005: four Lapland rivers flooded together.

The March snow warned us.



Rivers flood together, not one at a time

Springs when ≥4 of 14 rivers flood

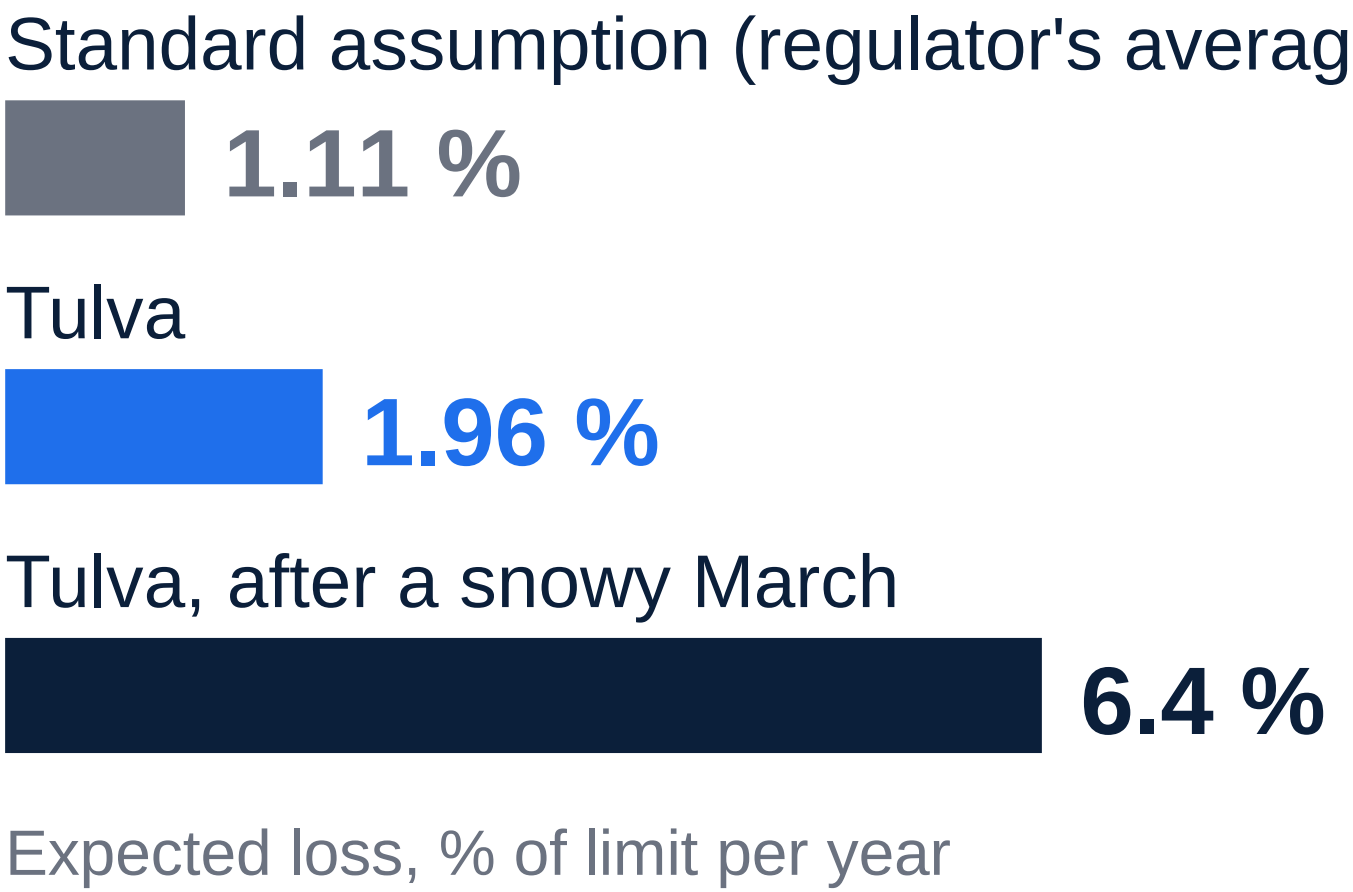


≈4×

as often as chance

Cover is bought in autumn, before the snow.

Tulva Index: pays if ≥2 northern rivers flood



×3

expected loss after a snowy March

Tulva

Joint risk | Snow outlook | Index cover | How it works | Under the hood | Year replay

1 · JOINT RISK

Finland's spring floods don't arrive one at a time.

The 14 river flood-risk areas fall into three clusters that share a snowpack. Lines join areas whose spring peaks move together (Spearman ρ ≥ 0.6, 1961–2025).

2.8×

per river pair: as often as chance, two rivers of one cluster both over 1-in-10 in the same spring (pairs) EXACT

10 of 65

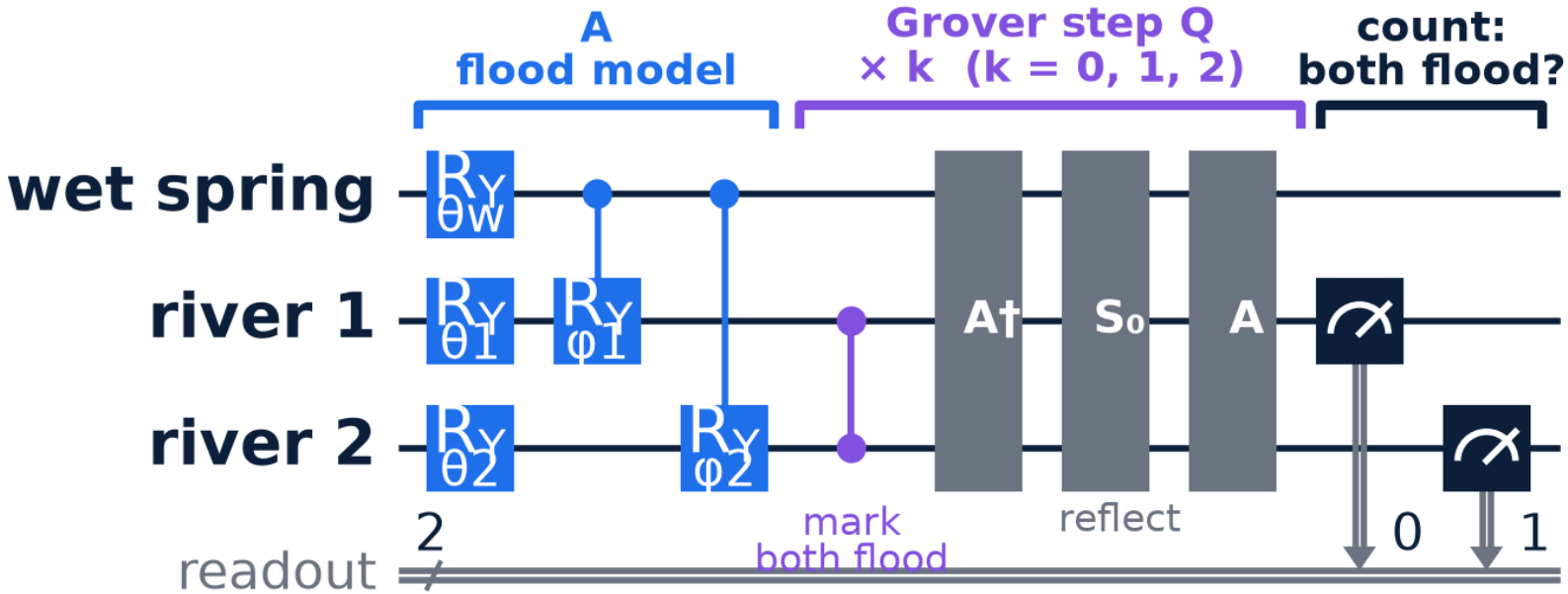
per spring: springs with ≥4 of 14 areas over their 1-in-10 level (3.9× chance) EXACT

- North · 5 areas · 2.9× joint floods Pudasjärvi, Ivalo, Kittilä, Rovaniemi, Tornio
- Ostrobothnia · 6 areas · 3.2× joint floods Lapväärtti, Ilmajoki–Seinäjoki, Laihia, Ylistaro–Koivulahti, Alavieska–Ylivieska, Lower Pyhäjoki
- Satakunta–SE · 3 areas · 1.3× joint floods Pori, Huittinen, Lower Kymijoki



Our app: lines join rivers that flood together

Our circuit: quantum amplitude estimation



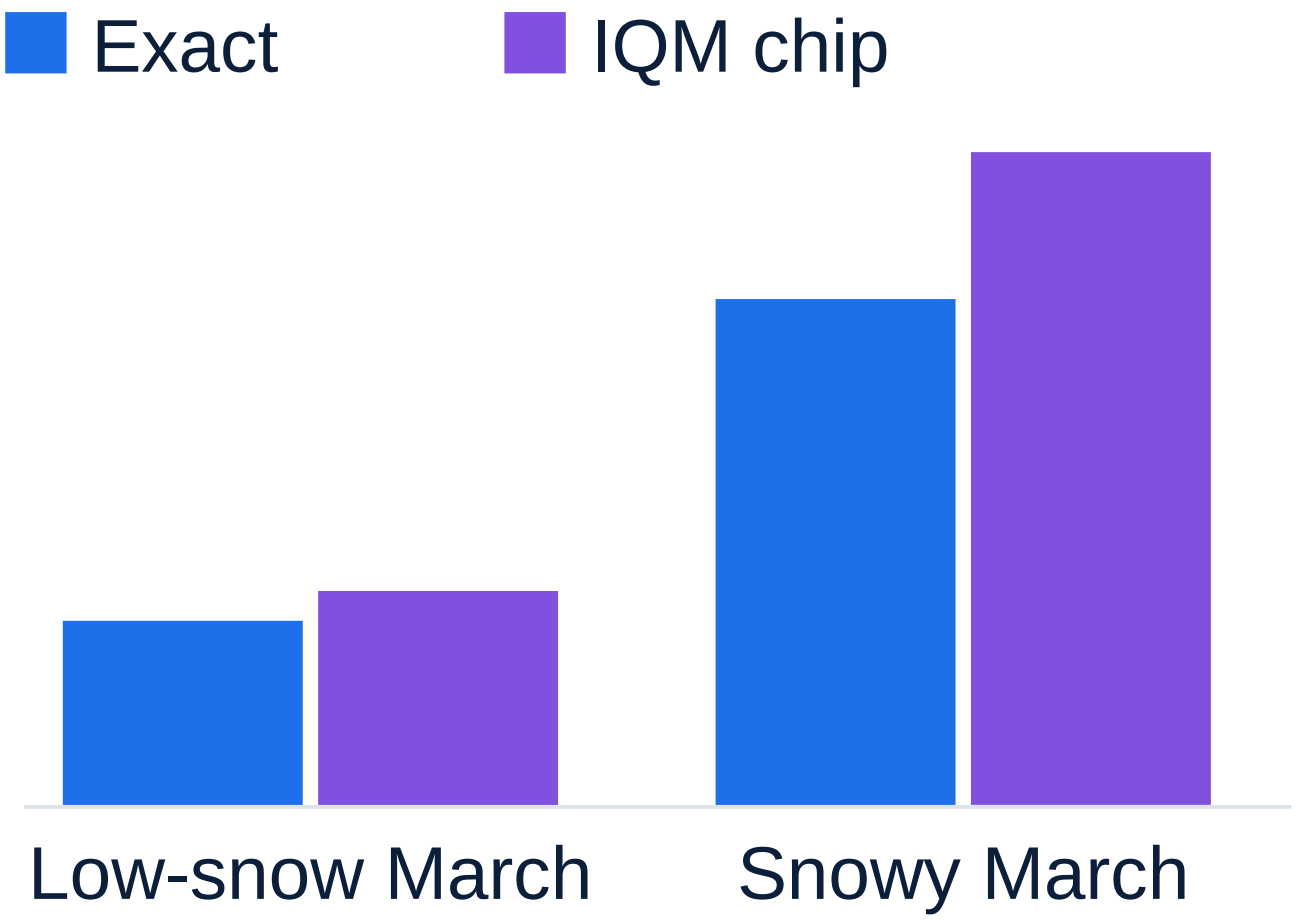
Qubits = wet spring + rivers. Gates = our flood model.

Why quantum? 2× accuracy: sampling needs 4× the runs, QAE 2×.

Perfect simulator: exact 0.125 %, ≈2.5× tighter than Monte Carlo

MLAE, Grover steps k = 0, 1, 2 · simulator ±0.011 vs Monte Carlo ±0.027 pp, 16,000 runs each · 2× vs 4× = ideal scaling

On real chips, the snow effect still shows



Snowy vs low-snow March

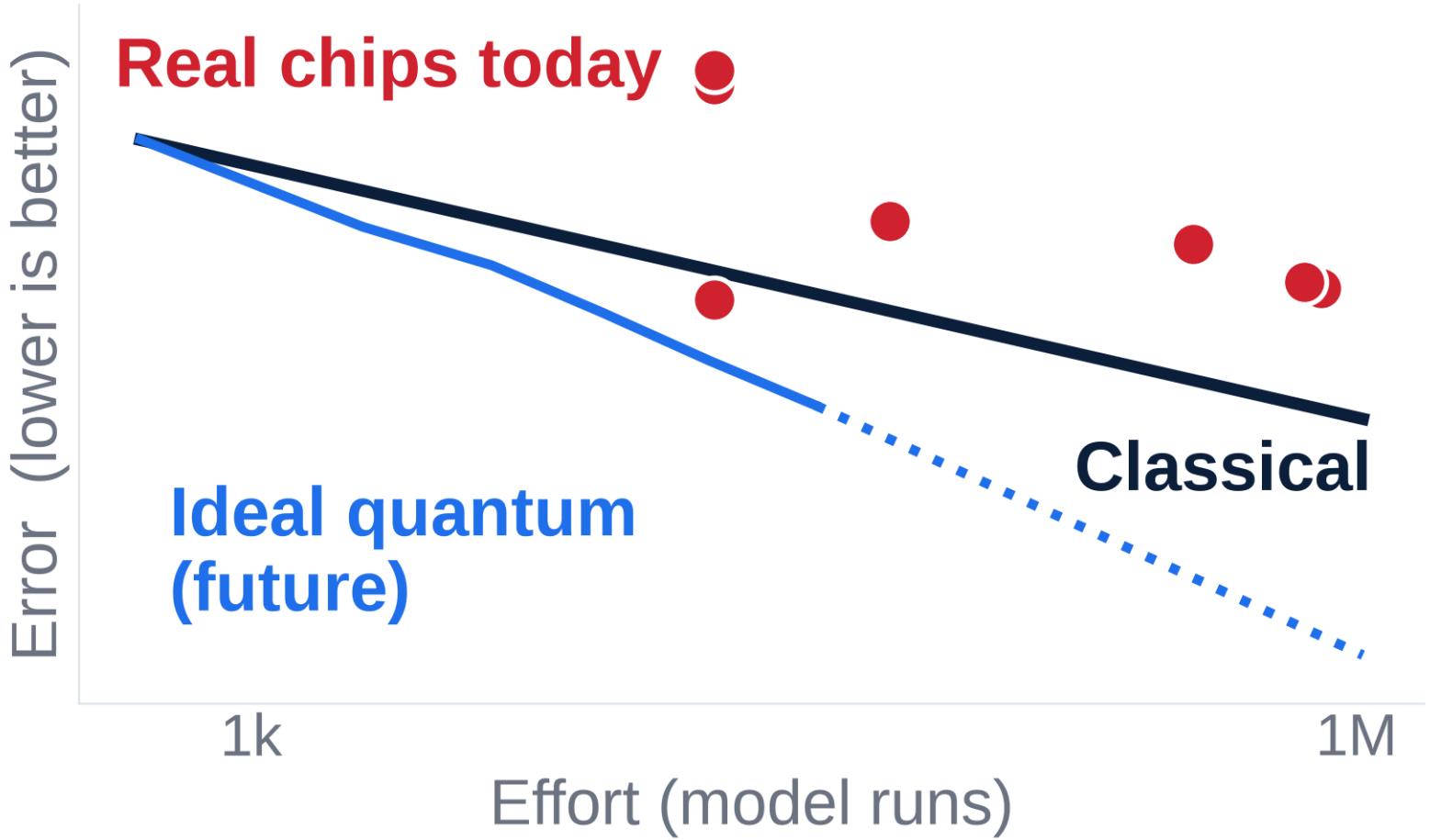
3.05× on chip

2.75× exact

Ran on

IBM · IQM · VTT Q50

Classical still wins today



Rare 1-in-50 floods drown in noise.

Prices are computed classically.

Who pays



€25k pilot, then €75k / yr

Year 3: €0.2–0.45M / yr

3–6 Finnish insurers, then the Nordics

Ask: one insurer's flood claims, 8 weeks.

Built this weekend

- SYKE data pipeline
- Joint flood model
- Index pricing
- Live app
- Runs on 4 quantum chips

Next March, the snow will warn us again.

This time, it will have a price.

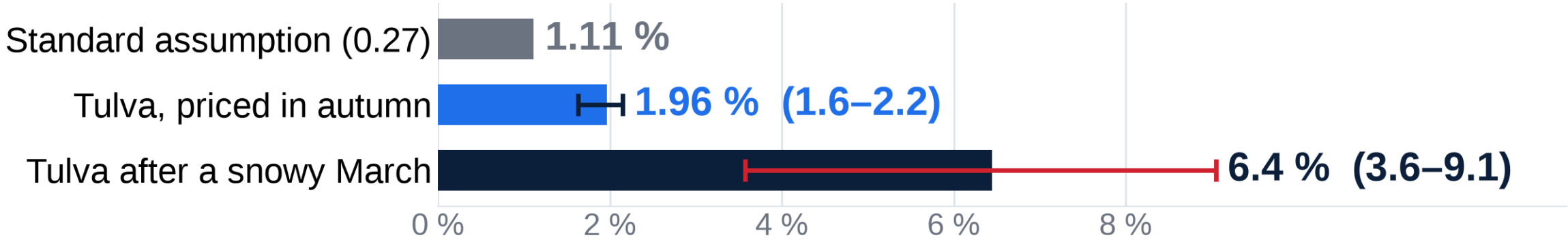
Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026

BACKUP · Q&A ONLY

Product and pricing in detail

EXPECTED LOSS, % OF THE LIMIT PER YEAR (YEAR-BOOTSTRAP 90 % INTERVALS)



- Index: pays the full limit if ≥2 of 5 northern rivers pass their 1-in-50 level (a staged payout is only a variant). Priced in autumn; in March the insurer can buy top-up cover and staff claims in the regions at risk.
- Tulva 1.96 % (1.6–2.2 %) vs 1.11 % for the standard assumption (a model with one 0.27 correlation) = 1.77× (≈1.8×). 0.27 = mean of the 2027 Finnish 19-zone flood matrix, Del. Reg. (EU) 2026/269. ≈5× only vs independence (0.38 %).
- Snowy (top-15 %) March vs autumn (normal): 6.4 % (3.6–9.1 %) vs 1.96 % = 3.3× (90 %: 2.1–4.4×).
- Market price = expected loss × 2.5–3.5: 4.9–6.9 % rate on line ≈ €49–69k a year per €1M of cover.
- Slide 1: 'flooded' = flow over each gauge's 1-in-10 level; North's 1 March snow in its top 15 % (in-sample). Slide 2: 10 of 65 springs (15.4 %, 90 %: 9–24 %) had ≥4 of 14 areas over 1-in-10; chance on the same gauge-years 3.9 % (3.9×), 4.4 % if all gauges reported (3.5×). SYKE 1961–2025 (+ SMHI Torne).

BACKUP · Q&A ONLY

Basis risk and business in detail

Basis risk (1-in-10 back-test). Paid in loss springs 1966, 2005, 2018; missed Ostrobothnia 2012–13 and 2024 and Lapland 2020; since 2005 also paid in 3 North springs with no recorded loss (2015, 2023, 2024). At the 1-in-50 product level: paid 1966 and 2005 only.

Why now. From 2027 the Solvency II standard formula fixes flood correlations that ignore the snow (19-zone matrix inside Finland, mean ≈ 0.27; 0 % to SE/NO/DK; Del. Reg. (EU) 2026/269). Insurers with an approved internal model, like If, can use a snow-aware joint-flood number.

Money flows. Insurer → reinsurer: premium ≈ €0.5–0.7M a year per €10M North layer (model × 2.5–3.5 market multiple). Insurer → Tulva: licence, €25k pilot, then ~€75k a year (assumption). The reinsurer pays Tulva nothing.

Year 3. 3–6 insurers × €75k ≈ €0.2–0.45M a year (assumption); Nordic snowmelt upside unsized (assumption). Buyers: Finnish home insurers (OP, If, LähiTapiola, …); no pilot agreement yet.

Market size. Losses per event ≈ €5–20M; parametric insurance market $3.5bn to $18–21bn, 12–14 %/yr growth (business case, outside sources, quoted as a range).

Ask. One insurer's flood claims for 8 weeks → basis-risk map + firm quote for 1 Jan 2028.

BACKUP · Q&A ONLY

Hardware runs in detail

Chip	Circuit	Exact	Raw	Calibrated (90 % CI)	Job
IBM Marrakesh	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.262 %	**0.149 % (0.140–0.157)**	db3tddsvf2bc73ctoprg
IQM Garnet	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.368 %	**0.164 % (0.147–0.180)**	01a11cc3-7f6c…
IQM Emerald	Snowy / low-snow ratio, same pair, k = 0–2	2.75×	1.3×	**3.05× (2.6–3.6)**	01a11fa8-65a9…
IQM Emerald	14 rivers ≥4 over 1-in-10, 18 q / 44 CZ	10.9 %	–	**13.8 % (readout-corr.)**	01a11cd4-da38…
IQM Emerald	14 rivers ≥2 over 1-in-50, same circuit	4.8 %	–	**16.0 % (fails)**	01a11f9b-c62a…
IQM Emerald	Control: independent circuit, ≥4 over 1-in-10	4.4 %	–	**9.7 % (noise fattens tails)**	01a120f5-ece7…
VTT Q50	Same pair, 3 repeats: not stable	0.125 %	1.48–1.58 %	**0.47 / 0.53 / 0.11 %**	7d20d8ac / a6fb9d52 / 381179fb
VTT Q50	14 rivers ≥4 over 1-in-10, 18 q / 49 CZ	10.9 %	34.2 %	**31.2 % (readout-corr.)**	9cdb0158-7574…

Method: calibrated sampling + shallow amplitude estimation (noise-aware MLAE, Grover steps k = 0–2, two known-answer calibration circuits in the same job; bootstrap 90 % intervals). Pair = two rivers with identical North-cluster loadings and 1-in-50 trigger. Snowy vs normal on the same pair: 1.51× chip, 1.55× exact. Simplified 14-river model = two-level version, 18 qubits, sampling only. IBM / Garnet intervals exclude exact (+19 % / +31 % bias). Classical Monte Carlo at the same budget is more precise (±0.004 pp).

Signal kept at k = 1 in the runs shown: Emerald 0.56–0.59, IBM / Garnet 0.77–0.78, VTT Q50 0.12–0.19 (earlier runs 0.65–0.88): the estimate is mostly the calibrated k = 0 circuit. Q50's zero-calibration offset moved 0.002 → 0.006 between repeats (the 5× spread). Not shown: first runs IBM 0.146 % (k = 0–3), Garnet 0.073 % (1 copy).

Resources: SYKE open data · IBM Marrakesh · IQM Garnet & Emerald (Resonance) · VTT Q50 (LUMI) · Qrisp · Qiskit · AI coding agents.